

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP.

#XUV
ORGANIC CHEMISTRY:
SOME BASIC PRINCIPLES
AND TECHNIQUES

LECTURE-10

CHEMISTRY

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Topics To Covered

1

Electron Displacement Effects

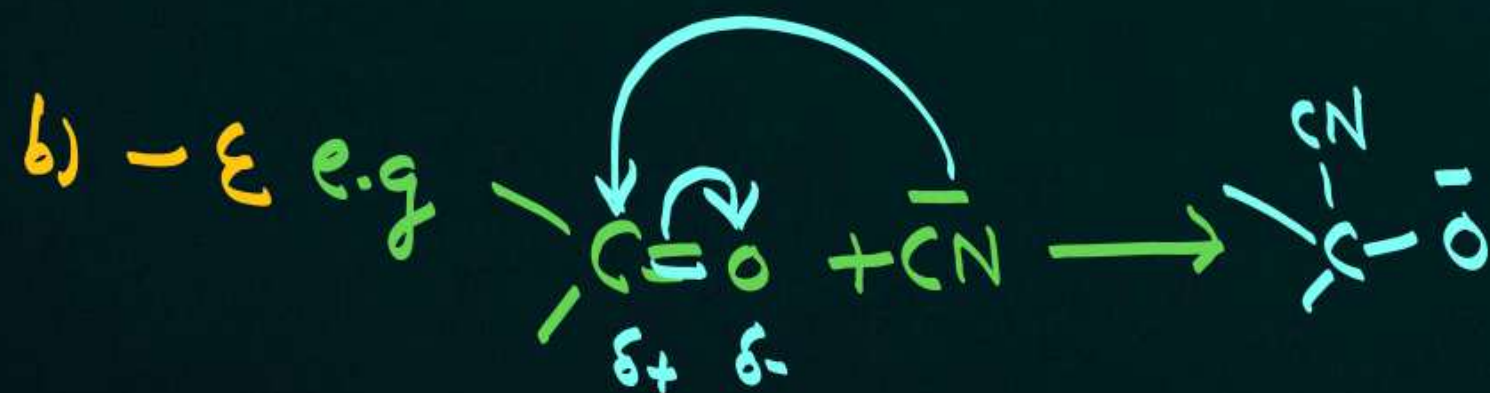
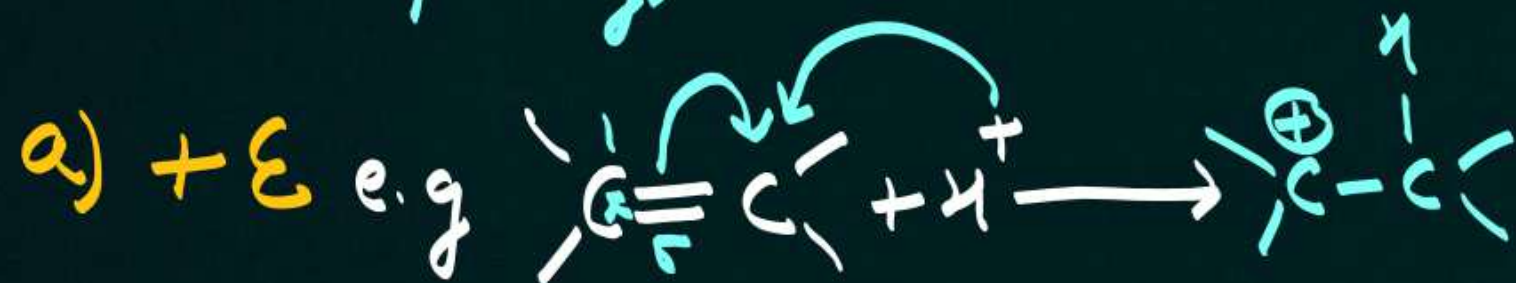
(Hyperconjugation & Resonance Effect)

REVISION.

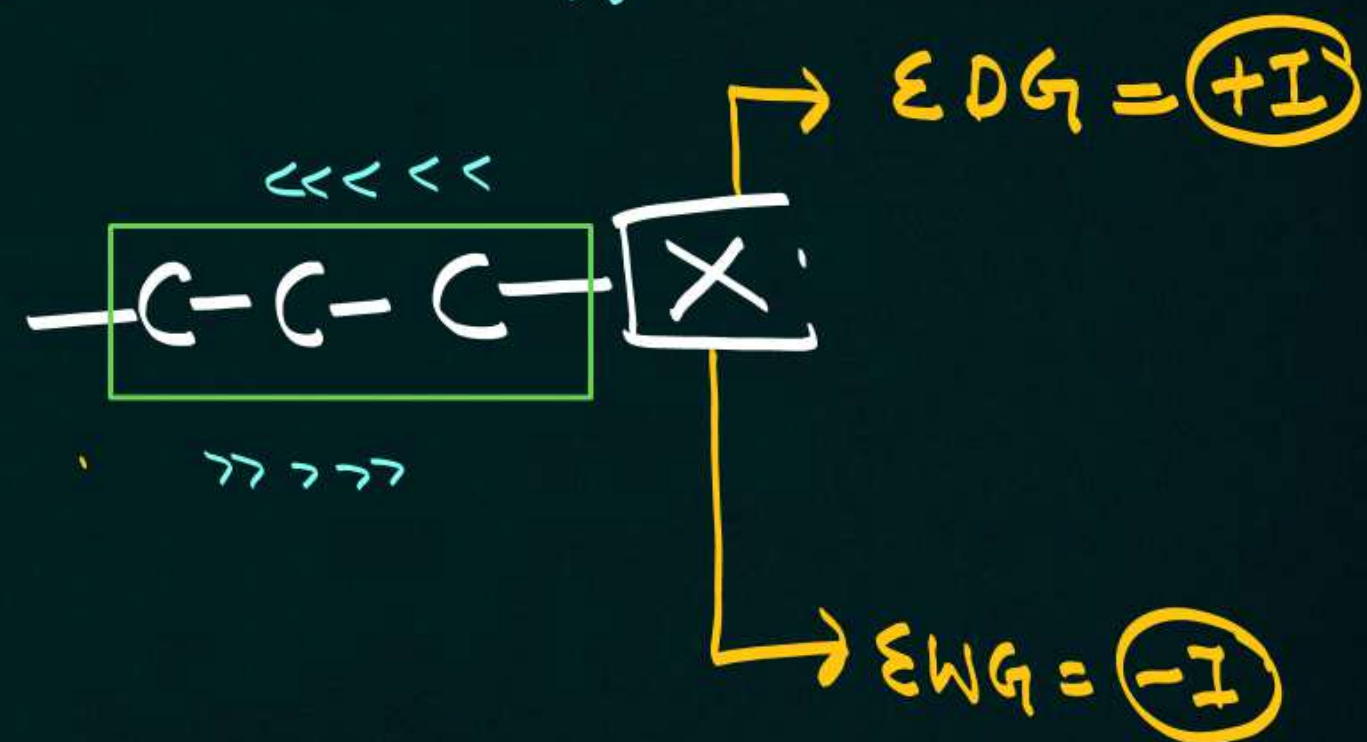
tert butyl > Iso-pro > ethyl > methyl.

$\bar{e}$ displacement effects.

1) Electromeric (πe)
(Temporary)



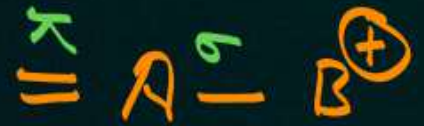
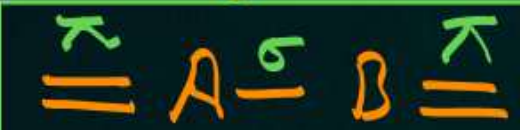
2) Inductive effect



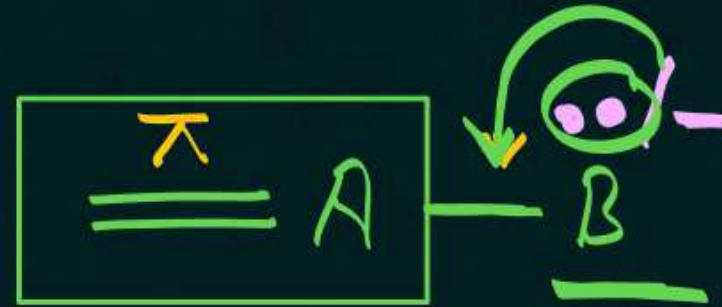
$CF_3 > NO_2 > CN > COOH > F > Cl > Br > I$

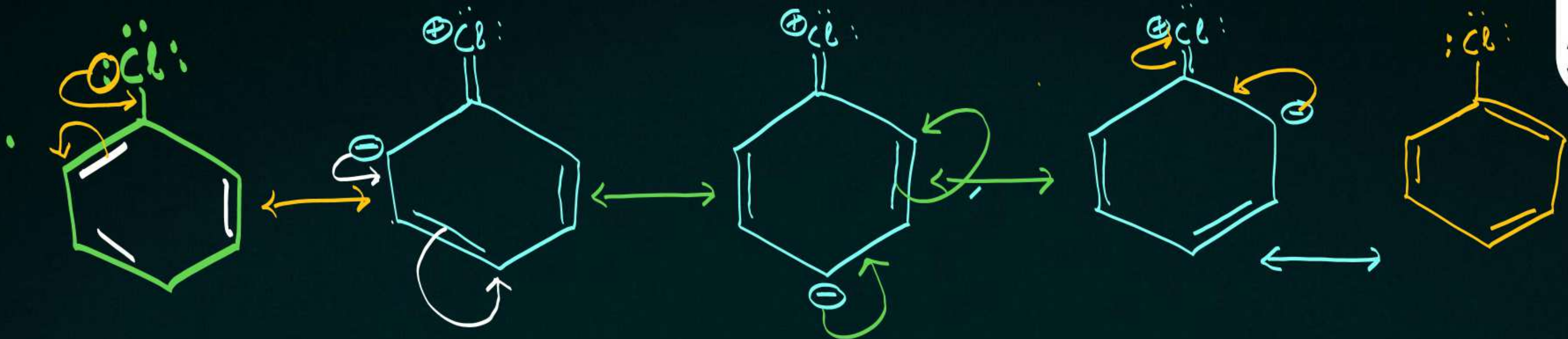
3) Resonance effect $\Rightarrow$

Polarity produced.



$\boxed{+R}$



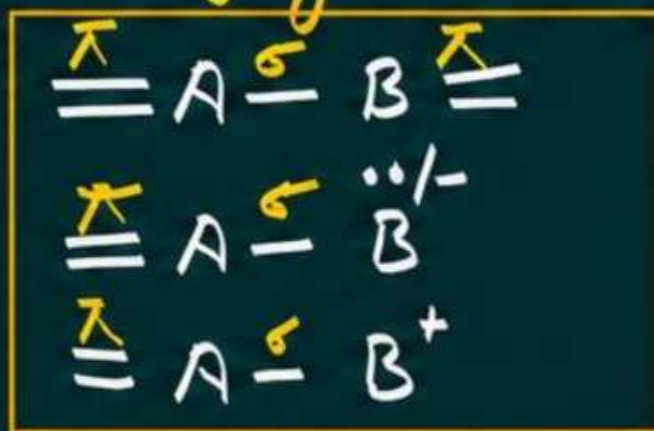


③ RESONANCE EFFECT → It is the polarity produced in a molecule by the interaction of two π bonds or one π bond and lone pair.

- This effect is produced in a conjugated systems.

- These are of two type
 - a) +R (positive Resonance) effect

- b) -R effect (Negative Resonance).

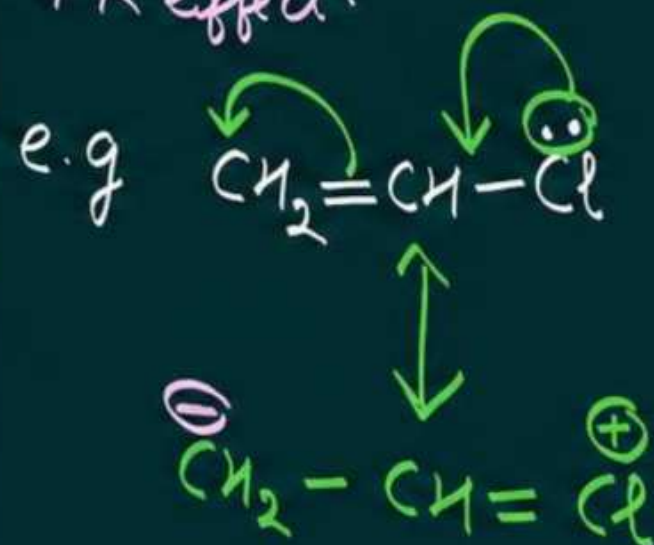


a) +R effect →

If the substituent has the tendency to donate electrons to the double bond or conjugated system, the effect is called positive resonance effect.

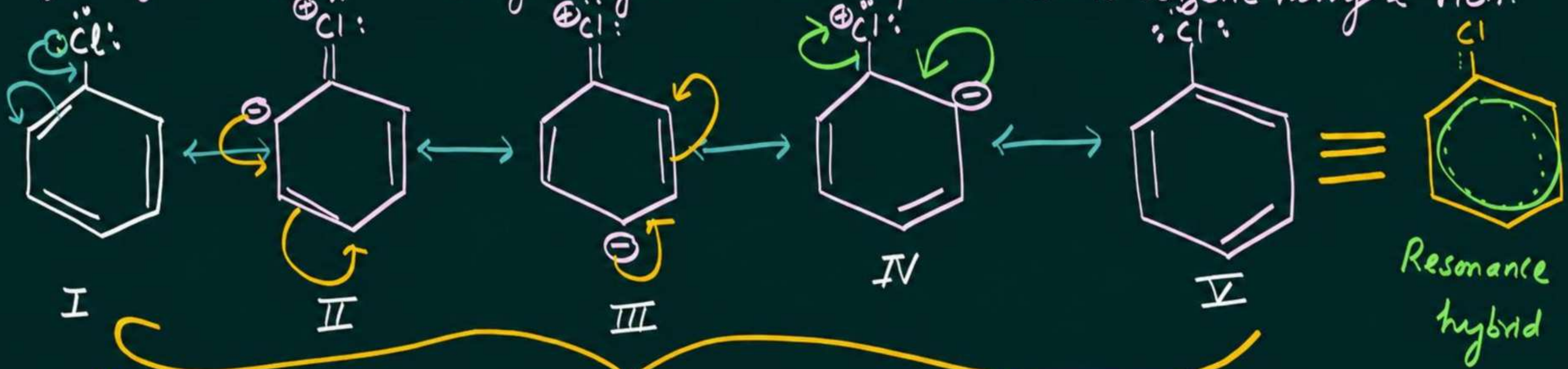
For example groups like.

$-\ddot{O}H$, $-\ddot{N}H_2$, $-\ddot{N}R_2$, $-\ddot{Cl}$, $\ddot{Br}$, $\ddot{I}$ etc show +R effect.



+R Group Ki Jaga pehchan,
Pehla hi atom par lone pair ka Nishan

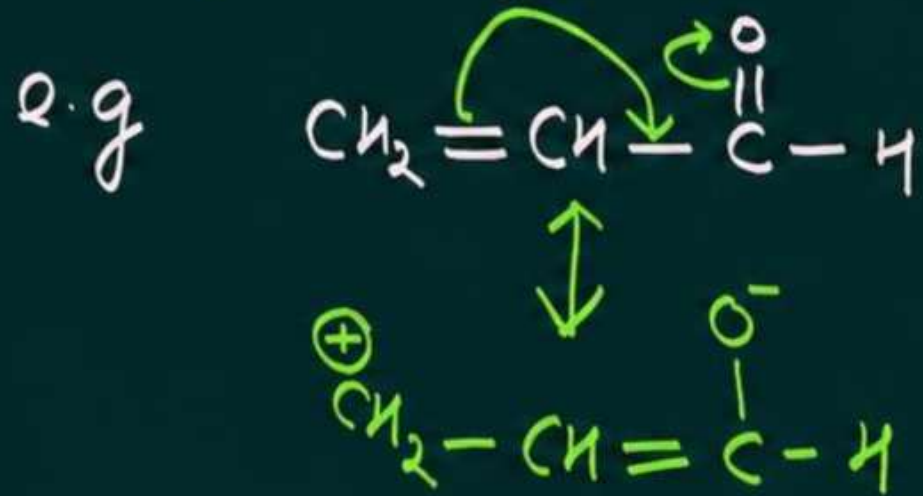
If +R group is present benzene ring, then +R Group makes the benzene ring $\bar{e}$ rich.



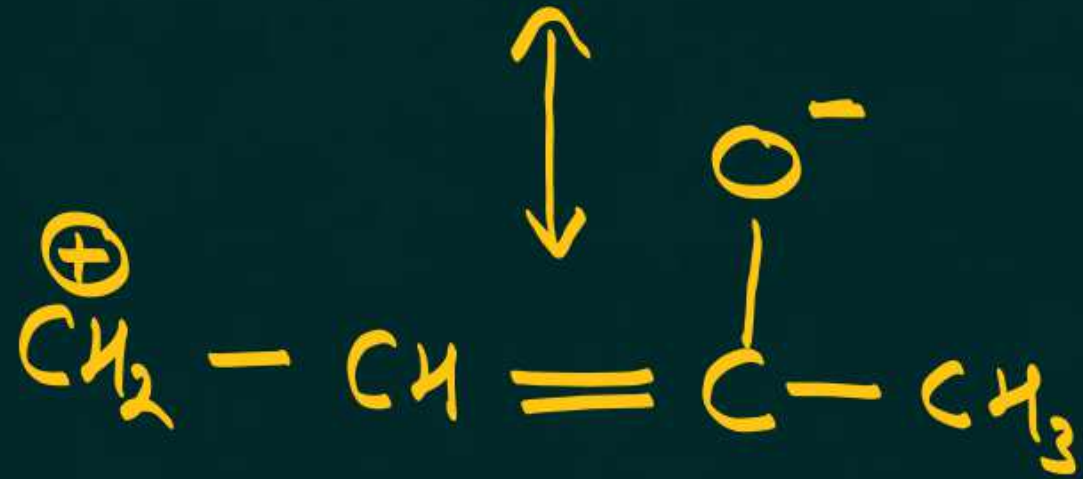
Due to +R effect $\bar{e}$ density increases on ortho & para position of benzene ring

b) -R effect $\rightarrow$ If a substituent attached on double bond or a conjugated system has tendency to withdraw electrons from double bond to itself, the effect is called -R effect.

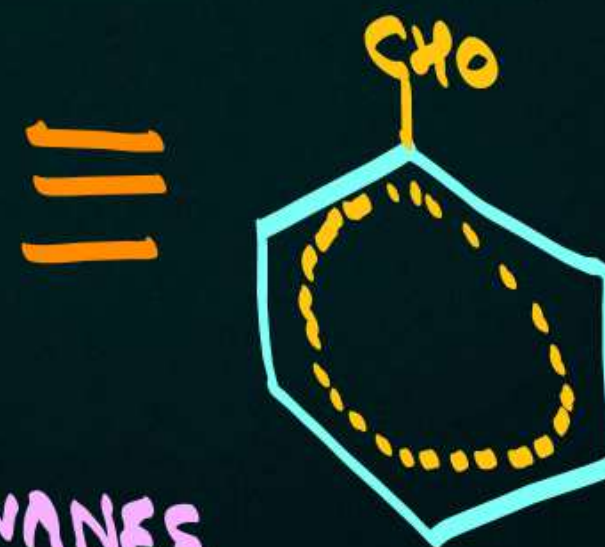
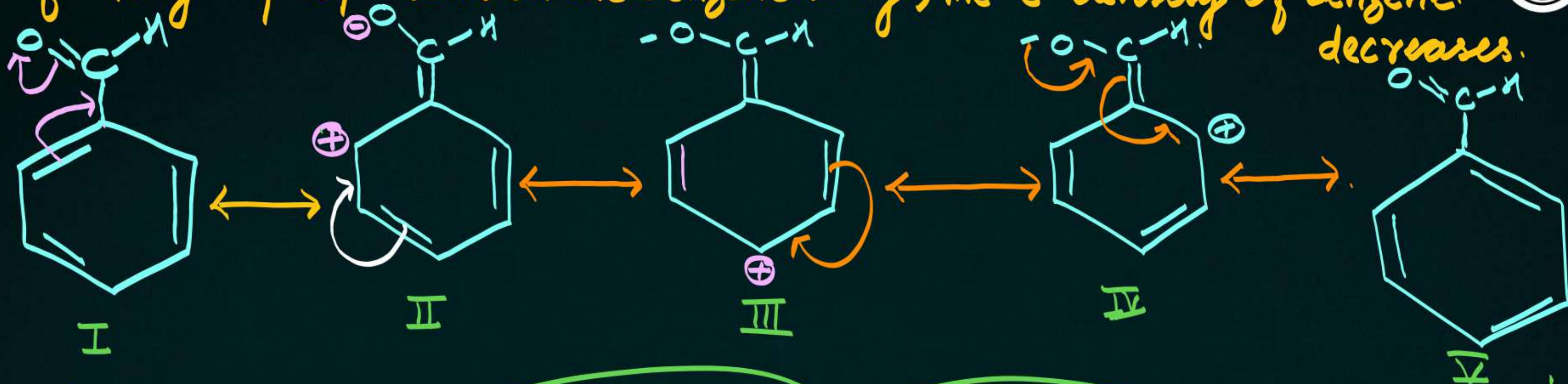
example. $\overset{\text{O}}{\parallel} \text{C}-\text{H}$, $\overset{\text{O}}{\parallel} \text{C}-$, $\overset{\text{O}}{\parallel} \text{C}-\text{OR}$, $\overset{\text{O}}{\parallel} \text{C}-\text{NH}_2$ etc. $\text{C}\equiv\text{N}$



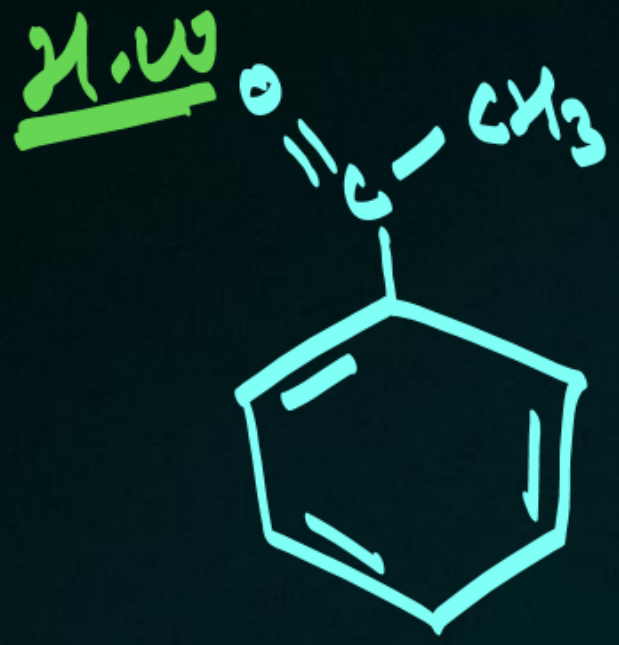
-R-group ki
xya Pehchan
Pehla hi atom
pe double or MB
ka dishan.



If -R-group is present on the benzene ring, the $\bar{e}$ density of benzene decreases.



RESONANCE HYBRID.



Q. Identify the group showing +R effect.



+R $\rightarrow$ Pehla Atom pe lone pair

-R $\rightarrow$ " " " MB Ka

Nishan.

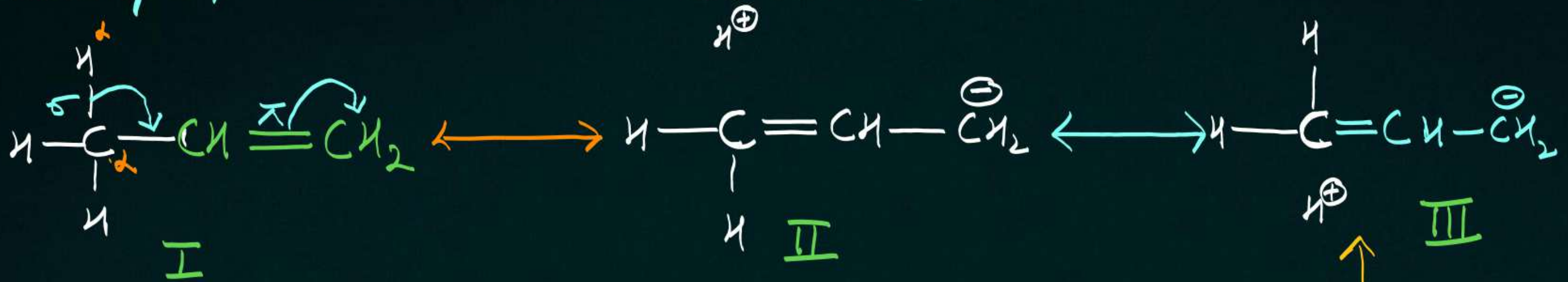
4). HYPERCONJUGATION $\rightarrow$

When in a organic molecule if C-H is attached with unsaturated system.

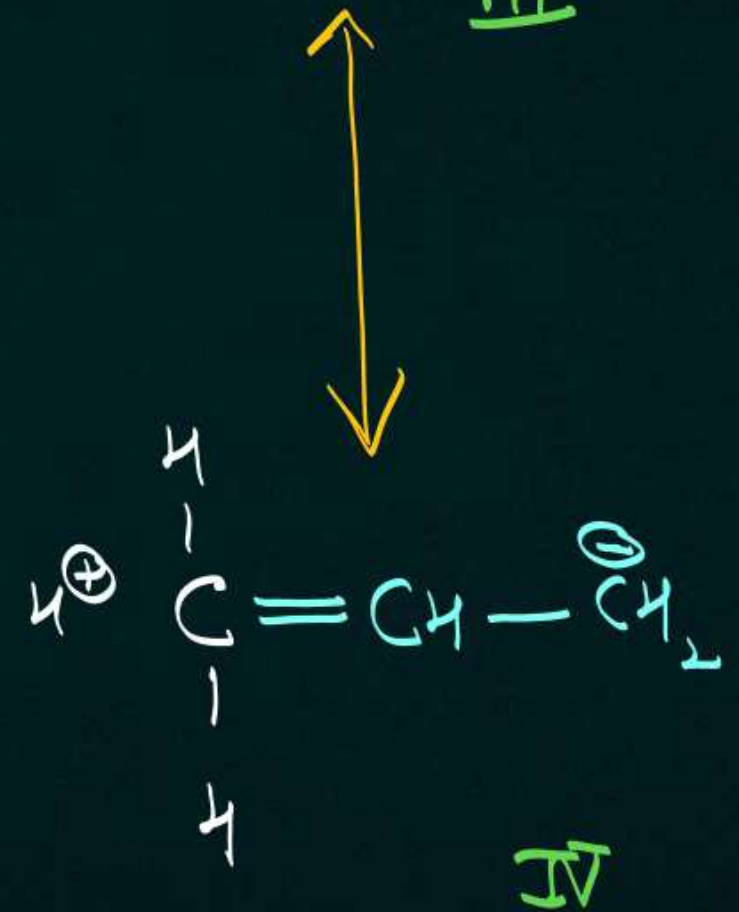
(multiple bond or benzene ring), the σ electrons of C-H bond starts interacting with π e's of multiple bond, This interaction is called. hyperconjugation.

This effect was given by Baker-Nathan.
i.e why it is also known as Baker Nathan effect. OR No bond Resonance.

Example a) when C-H is attached with C=C.



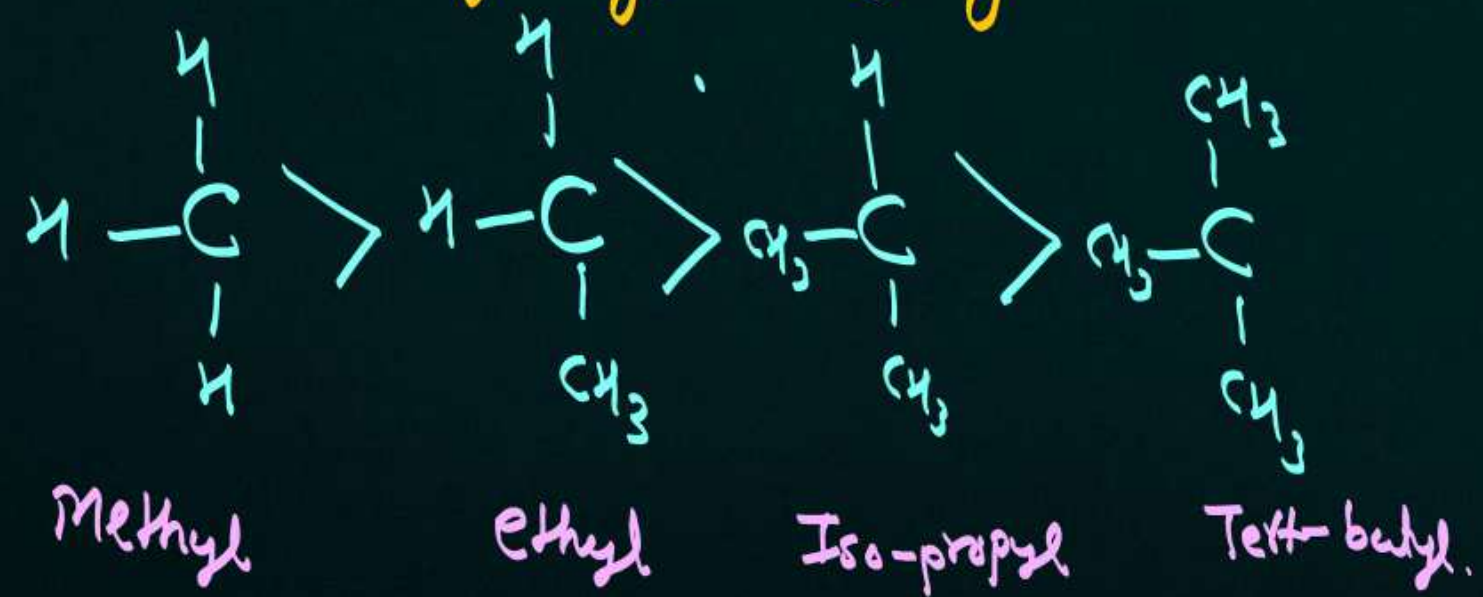
No. of hyperconjugative structure
(No. of α -H + 1)



b). When $-C-X$ is attached with carbon having +ve charge.



Order of hyperconjugation

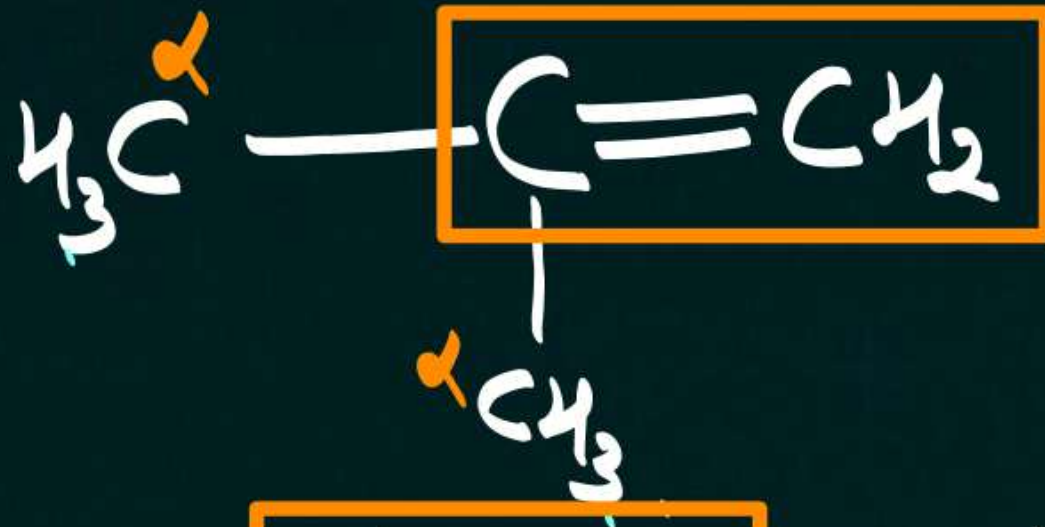


1) NUMAAN WANI

2) MEER FAISAL

3) Para COMMANDO TOUFEEQ.

Q. How many hyperconjugative structure are possible for following



No. of hyperconjugative str.
(No. of α -H + 1)
 $6 + 1 = \boxed{7}$

a) 4

b) 6

c) 7

d) 5

Thank
You